Quantum Algorithms Miniproject - Quantum Extreme Learning Machines

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1 Introduction

Traditional classical neural network models are trained by iteratively adjusting all their weights through back-propagation which is computationally expensive due to the individually calculated updates of each weight. Extreme Learning Machines (ELMs) were introduced as an alternative to traditional machine learning algorithms, to remedy this issue [1]. These algorithms are designed for feed-forward neural networks with a single hidden layer with the weights between the input and hidden layers being randomly initialised and remaining fixed during training. This eliminates the need for iterative weight adjustment. Instead, training focuses on optimising only the output weights of the model, by solving a system of linear equations to determine the optimal values. This process transforms training into a convex optimisation problem, significantly reducing computational complexity and training time compared to gradient-based training approaches. ELMs can be applied for several different tasks, such as regression, classification and clustering, among numerous others [2].

With the emergence of Quantum Machine Learning, much research has been done regarding the quantisation of ELMs. Quantum Extreme Learning Machines (QELMs) are appealing due to their ability to encode input data into a high-dimensional feature space through reservoir dynamics, as well as their ability to process quantum data. In addition, this remains a convex optimisation problem, ensuring models remain trainable through noise from quantum systems. Using this framework, the computational costs of quantum machine learning models could be significantly reduced. Here, an analysis on the expressivity and richness of QELMs is performed. Furthermore, the potential of QELMs is investigated by evaluating their performance on a Fourier series reconstruction task for several different target functions. To study the effect of the richness of the QELM on performance, several encoding schemes and reservoir dynamics are experimented with. Finally, a QELM application on a classification task is provided as well, showing the versatility of the algorithm by benchmarking its performance on the Fashion-MNIST dataset.

2 Background

2.1 QELM Framework

QELMs are quantum-classical hybrid models that retain the principles of classical-only ELMs. A QELM comprises a Parametrised Quantum Circuit (PQC) and a classical regression model. The PQC has a set of input nodes, known as "accessible" qubits and a set of hidden nodes, or "hidden" qubits, much like a traditional neural network. The circuit transforms input data by combining an encoding scheme on the accessible qubits with a set of reservoir dynamics on both the accessible and hidden qubits. This results in a quantum state that can be measured on a set of M unique observables $\{O_k\}_{k=1}^{M}$, of which the expectation values $\langle O_k \rangle$ provide an encoding vector. Then, the classical regression model is trained to predict accurate labels based on this vector. Like a classical ELM, the PQC is fixed during training, and only the regression model requires tuning.

The maximum size of the encoding vector scales exponentially with the size of the PQC. For instance, if the set of single-qubit observables is limited to $\sigma_x, \sigma_y, \sigma_z$ and I, the number of unique system observables is 4^{n_q} for n_q qubits. This ensures that a high-dimensional encoding can be achieved without notable computation costs. The fixed encoding architecture of the QELM allows for significantly faster training times compared to conventional learning algorithms while providing similar results. Additionally, QELMs can be trivially adapted for different tasks, by adjusting the regression model to output labels of a specific model type.

2.2 Fourier representation analysis

As mentioned previously, the PQC outputs a vector of expectation values of a set of observables. The expectation value of a single observable M is defined as $\langle \psi | M | \psi \rangle$, with $|\psi\rangle = U(x)|0\rangle$ being the output state of the PQC, obtained from applying a parametrised unitary $U(x)$ on initial state $|0\rangle$. $U(x)$ can be decomposed into L layers, each consisting of a trainable gate W followed by a parametrised encoding gate $S(x)$, which applies single-qubit rotations to the full circuit. Single-qubit rotations are governed by their time evolution, according to a given Hamiltonian generator H. Thus, the parametrised unitary $U(x)$ can be written as follows:

$$S(x) = \exp(iHx) \tag{1}$$

$$U(x) = W^{(L+1)}S(x)W^{(L)}....W^{(2)}S(x)W^{(1)} \tag{2}$$

Since W is an arbitrary circuit, the Hamiltonian can be diagonalised to the form $H = V^{\dagger}H'V$ without loss of generality by including V and $V^{\dagger}$ in W. Due to the fact that the eigenvalues $\lambda_1, ..., \lambda_d$ of H' are on its diagonal, the output state $|\psi\rangle$ is described by

$$|\psi\rangle = U(x)|0\rangle = \sum_{J\in[d]^L} \exp(-i\Lambda_J x) W^{(L+1)}_{ij_L} ... W^{(2)}_{j_2 j_1} W^{(1)}_{j_1 1} \tag{3}$$

with J as a multi-index for L integers $\{j_1,..,j_L\}$ between 1, d and Λ_J being the sum of eigenvalues $\lambda_{j_1},...,\lambda_{j_L}$. Substituting this expression into the definition of the expectation value results in the expression $f(x)$ for the full circuit:

$$f(x) = \sum_{K,J\in[d]^L} \exp(i(\Lambda_K - \Lambda_J)x) a_{K,J} \tag{4}$$

$$a_{K,J} = \sum_{i,i'} (W^*)^{(1)}_{1k_1} ... (W^*)^{(L+1)}_{k_L i} \times M_{i,i'} \times W^{(L+1)}_{i' j_L} ... W^{(1)}_{j_1 1} \tag{5}$$

The unique set of frequencies Ω give the frequency spectrum of the PQC. Summing the $a_{K,J}$ that contribute to the same frequency $\omega = \Lambda_K - \Lambda_J$ provides the coefficients c_ω for the Fourier representation of the PQC:

$$f(x) = \sum_{\omega\in\Omega} c_\omega \exp(i\omega x) \tag{6}$$

$$\Omega = \{\Lambda_K - \Lambda_J, K, J \in [d]^L\} = \{(\lambda_{k_1} - \lambda_{j_1}) + ... + (\lambda_{k_L} - \lambda_{j_L}) : k_i, j_i = 0, 1\} \tag{7}$$

Note that Ω is solely dependent on the eigenvalues of the Hamiltonian, and c_ω is a result of $|0\rangle, U(x)$ and M. In the case of QELMs, the expectation values $\langle O_k \rangle_x$ obtained from the PQC are further processed by the regression model. It predicts a label through a weighted sum of the output vector, given weights $[\eta_1, ..., \eta_M]^T$:

$$f_\eta(x) = \sum_{k=1}^{M} \eta_k \langle O_k \rangle_x \tag{8}$$

Denoting the Fourier coefficients of a specific observable O_k as a^k_ω, the Fourier coefficients for a specific frequency can be calculated with $b_\omega = \sum_{k=1}^{M} \eta_k a^k_\omega$. The Fourier representation of the QELM can then be described as

$$f_\eta(x) = \sum_{\omega\in\Omega} b_\omega \exp(i\omega x) \tag{9}$$

2.3 Data encoding

A classical input vector x can be encoded into a QELM through a parametrised unitary $U(x)$. Often, this encoding is achieved by applying a set of single-qubit Pauli rotation gates on the accessible qubits. Similar to Equation 1, the rotation of qubit k is governed by its time evolution, according to a given Hamiltonian generator H_k:

$$U_k(x) = \exp(iH_k x) \tag{10}$$

Note that qubits can have a unique Hamiltonian H_k applied to them, causing different rotations in each qubit. Depending on the set of H_k used, the encoding scheme changes.

The most trivial rotation-based encoding method is known as Pauli re-uploading, defined as evolving all qubits through the same generator. Consequently, each qubit has the same rotation applied to it [3]. This limits the size of Ω since all H_k have the same eigenvalues. To increase the number of achievable frequencies, unique generators can be applied to each individual qubit. Optimally, the frequencies obtained from different Hamiltonians are unique, as this results in the largest size of Ω. This can be achieved by exponentially separating the eigenvalues of each Hamiltonian, which is referred to as exponential encoding [3].

The number of functions that a QELM model f_η can approximate, known as the Fourier-expressivity $F[f_\eta]$, differs between encoding schemes. The output of the QELM is computed as a linear combination of M observables, each being a linear combination of 4^{n_q} Pauli strings. An assumption can be made that M will always be smaller than 4^{n_o} in practice, due to the fact that exhaustively measuring every possible observable can result in added redundancy and reduce model efficiency. In addition to being upper-bounded by M, f_η can be decomposed into a Fourier series over all frequencies in Ω, as shown in Equation 9. The size of Ω is upper-bounded by the number of unique different values of $\Lambda_K - \Lambda_J$, which is 4^{n_a} since the encoding unitary has 2^{n_a} eigenvalues. Thus, the upper bound of the Fourier-expressivity for a QELM is described as follows:

$$F[f_\eta] \leq min\{M, |\Omega|\} \tag{11}$$

2.4 Reservoir Evolution

The role of the reservoir unitary is to ensure the model stays rich, meaning that a large proportion of frequencies have non-zero coefficients. As shown in Xiong *et al.* [3], a QELM without a reservoir sees a concentration of its non-zero coefficients to a very narrow array of frequencies, which severely limits the expressiveness of the model. The reservoirs found to be most effective by Xiong *et al.* [3] are Haar random and 1D chaotic Ising unitaries. These have a scrambling effect on the frequencies, leading to a low concentration of Fourier coefficients and a rich model that's able to represent a wider array of functions.

A Haar Random unitary can be obtained by applying a Haar measure to the set of all possible unitaries, which ensures that the sample distribution is uniform across the full set. A fully random unitary can expand encoded information across the full frequency spectrum with no concentration.

The Ising Hamiltonian models the coupling and interactions between different qubit spins. This model can be described as

$$H_{Ising} = J \sum^{n-1} i = 1 Z_i Z_{i+1} + B_z \sum_i^n Z_i + B_x \sum_i^n X_i \tag{12}$$

with Z_i and X_i as Pauli operators for qubit i and J, B_z and B_x being coupling constants. For a chaotic Ising model, $J = -1, B_x = 0.7$ and $B_z = 1.5$. The complexity of this model causes small encoding changes to result in vastly different outcomes, resulting in a frequency spectrum similar to those obtained from a Haar random unitary.

2.5 Limitations

While QELM is a promising framework, it suffers from several limitations; most notably, the exponential concentration of observables' expectation values. This means that the observables' expectation values can concentrate around an input-agnostic value. This concentration can be deterministic, where the values concentrate in an exponentially tight bound, or be probabilistic where the output expectation values have an exponentially small chance to deviate from this value.

In studies like this one, where the system is simulated, the exponential concentration of observables' expectation values is not as much of an issue. Thus, there is access to the exact expectation values directly. However, this is a serious threat to the practicality of QELMs, as it would cause users to need an exponential number of measurements to differentiate values. Although training would still be possible, this would severely limit the generalisation capabilities of the system.

3 Method

3.1 Target Functions

To find the effects of the QELM in different scenarios, several different target functions are simulated including a Sphere function (13), the toy problem from [3] (14) and a Gaussian function (15). In each of these functions, the values fluctuate significantly, with the toy function displaying an exponential number of frequencies, the Sphere a quadratic number, and the Gaussian maintaining a constant frequency count. The following equations were used:

$$f_{sphere}(x) = \sum_{i=1}^{\frac{6^2-1}{2}} \alpha_k^2 + x^2 + \beta_k^2 \tag{13}$$

$$f_{toy}(x) = \sum_{k=1}^{\frac{3^6-1}{2}} \alpha_k \cos(kx) + \beta_k \sin(kx) \tag{14}$$

$$f_{GP}(x) = \sigma^2 \exp\left(-\frac{1}{2}\frac{|x-\mu|^2}{l^2}\right) \tag{15}$$

With $\alpha_k \sim \mathscr{U}(-1,1)$ and $\beta_k \sim \mathscr{U}(-1,1)$, and with lengthscale = $\sqrt{2}$. To see the effects of QELM on each of the target functions the average was taken over 5 runs, each of which had 5000 iterations. For testing, the same encoding and reservoir (exponential and haar random) were used for each target function and each run had a circuit with 6 accessible qubits and 0 hidden qubits.

The data points that are fed into the functions are 5000 equidistant points in the range $[0, 2\pi]$. Of these, 30% are set aside to test the model.

3.2 Preprocessing

Preprocessing was only used in the model's classification benchmark. This is mainly for computation reasons as simulating a system that has 28×28 accessible qubits was not practical. This also allows you to see the influence that preprocessing can have on the number of frequencies needed to model the problem. One can assume that Fashion-MNIST would require at least 28×28 frequencies to be modelled properly. However, by preprocessing each image it is possible to reduce the number of frequencies needed to model the function closely. To preprocess the Fashion-MNIST data, each input image was transformed into two scalars. The first is a combination of the centroids of intensity defined by $\sqrt{x_c + y_c}$ where x_c and y_c are the weighted averages of the intensity in the corresponding axes. The second scalar used is Shannon entropy, which follows $S = -\sum_{k=0}^{255} p_k * \log(p_k)$ where p_k is the frequency of pixel value k.

3.3 Encoding experiments

The effect of three different encoding unitaries are investigated in the following experiments. The first is the exponential encoding, presented by Xiong *et al.* [3], which follows Equation 16. This encoding allows for frequencies from $\frac{-3^{n_a}-1}{2}$ to $\frac{3^{n_a}-1}{2}$, for a total of $1 + \frac{3^{n_a}-1}{2}$ achievable frequencies, as given by Equation 7. This encoding thus allows for the number of achievable frequencies to be exponential in n_a. This encoding is the only one used for classification, as it takes vector inputs and k resets after processing each part of the vector (the original paper did not detail k's role for vector inputs).

$$H_k = \frac{1}{2} 3^{k-1} \sigma_z \tag{16}$$

The second encoding used here also comes from Xiong *et al.* [3] and is called Pauli re-uploading. In this encoding scheme, all generators are the same: $H_k = \sigma_z/2$. Thus the achievable frequencies scale linearly in the number of accessible qubits, reaching $n_a + 1$ non-negative frequencies.

The third encoding used here is an attempt to reach a scaling between Pauli reuploading and exponential encoding. Here, it is referred to as quadratic encoding. It follows $H_k = k^2 * \sigma_z/2$. The achievable frequencies of this model scale quadratically in the number of accessible qubits.

As the role of the reservoir was already investigated by Xiong *et al.* [3], this paper compares the performance of their two best-performing unitaries; the Haar random unitary, as well as the 1D Chaotic Ising model. The classification experiment only uses the Haar random unitary. Here, the Haar random reservoir was implemented using QR decomposition. As mentioned previously, thanks to the simulation aspect of these experiments and the small scale and depth, the exponential concentration of observables' expectation values is less of a concern. This allows us to use this reservoir and compare it to the second one that was implemented.

3.4 Observables

The observables of choice for all experiments were Pauli strings. This is in part due to their ease of implementation, as Pauli strings are well known and libraries already exist with ready-made implementations. They also appear to be a suitable choice thanks to their linear independence. This also replicates the work of Xiong *et al.* [3], who use Pauli strings as observables in all experiments. It is worth noting that this choice of observables may be unsuitable for a practical scenario, as the global measurements could lead to an exponential concentration of the observables' expectation values. However, in the experiments described here, the system has a small number of qubits (up to 6) and shallow encoding unitaries. This issue is also avoided by access to exact expectation values in one measurement.

3.5 QELM settings

In the model benchmark experiment, the toy problem presented in Xiong *et al.* [3] was reused, with the same setup of 6 accessible qubits and 0 hidden qubits. This ensures that the setup has been proven to work before and that the results given by different encodings can be compared on different problems to absolute values of the results obtained in the paper.

For the classification benchmark, a variety of settings were tested. Here, the results are reported on the setting that performed best: 4 accessible qubits (2 per data input element) and 2 hidden qubits. In both experiments, a fully controllable regime was used, meaning there are more observables than function frequencies. The results are reported using 4^{n_a} observables. For regression, the model uses a standard linear regression model. As performances for classification using a linear regression model were quite underwhelming, a ridge classifier was used.

4 Results

4.1 Regression

From the data in Table 4.1, it can be seen that the performance on the toy function exceeds the other functions by a large margin, being fit almost perfectly. The Gaussian is also fitted quite closely, since its output range is only 1 (compared to $\tilde{6}5$ for the toy function and $\tilde{6}76$ for

Target Functions	MSE (5000 samples)	MSE (200 samples)
$f_{toy}(x)$	$1.44*10^{-11} \pm 1.21*10^{-12}$	120.63 ± 6.03
$f_{sphere}(x)$	72.53 ± 1.45	$33,733.75 \pm 127.54$
$f_{gp}(x)$	$3.70*10^{-4} \pm 7.4*10^{-6}$	$0.059 \pm 1.18*10^{-3}$

Table 1: *Regression results of a QELM ($n_a = 6, n_h = 0$) evaluating the Fourier representation of several target functions for different dataset sizes. Results are averaged across 5 repetitions and reported with a 95% confidence interval.*

the sphere) the error is relatively larger than it seems. The gap between the toy function and the other functions might be due to the model over-smoothing the functions and its scaling fitting more poorly, therefore removing some details.

Table 2 shows the achieved error rate for QELMs trained with varying encoding schemes. Exponential encoding seems to significantly outperform Pauli re-uploading and has an even larger margin against quadratic encoding. Since the number of achievable frequencies scales exponentially instead of linearly (or quadratically), the model can obtain a frequency spectrum with a similar size to the target function. Consequently, the function can be reconstructed perfectly, achieving a near-perfect test error. Note that a perfect MSE would be 0, whereas the exponential encoding returns a value slightly above that. Given the difference in the order of magnitude compared to the results of other experiments, however, this may also be due to an error in floating-point precision.

Comparing reservoir dynamics to each other, the Haar random model slightly outperforms the 1D Chaotic Ising model. Both reservoirs are capable of generating complex Fourier spectra due to their dynamics being highly non-linear, resulting in the ability to capture a large variety of frequency patterns. As the 1D Chaotic Ising model has more structured interaction dynamics than the Haar random model, the resulting frequency spectra will be slightly more concentrated as well. This means that its Fourier spectra will be slightly less rich, resulting in a decrease in performance.

Encoding	Reservoir	MSE
Pauli Re-uploading	Haar Random	15.77 ± 1.33
Pauli Re-uploading	1D Chaotic Ising	16.91 ± 1.43
Quadratic Encoding	Haar Random	319.24 ± 5.32
Quadratic Encoding	1D Chaotic Ising	415.90 ± 6.42
Exponential Encoding	Haar Random	$1.44*10^{-11} \pm 1.21*10^{-12}$
Exponential Encoding	1D Chaotic Ising	$1.48*10^{-11} \pm 1.25*10^{-12}$

Table 2: *Regression results of a QELM ($n_a = 6, n_h = 0$) evaluating the Fourier representation of f_{synth}, for different encoding schemes. Results are averaged across 5 repetitions and reported with a 95% confidence interval.*

4.2 Classification Task

Over 5 runs on the entire fashion-MNIST dataset, the model achieved an accuracy of 48.51%. Over these runs the variance seemed quite low, the largest performance gap was approximately 0.5 points, indicating that the model is quite stable. As a multi-class ridge classifier is used, there is one set of richness coefficients per class (10 in this case). Thus, richness is computed from the average of these coefficients. As can be seen in Figure 1, most of the coefficients have non-zero values. Of all 2560 coefficients (256 observables * 10 classifiers), only 640 or 25% are null. This indicates that the reservoir is scrambling properly.

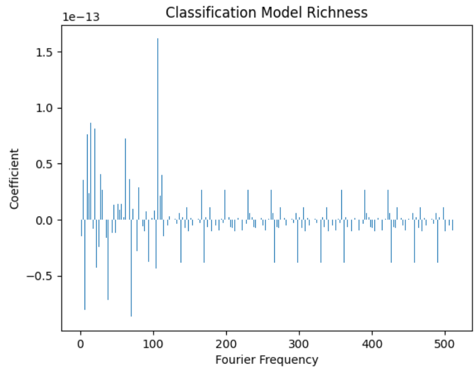

Figure 1: *Fourier coefficients plotted per frequency for a trained classification model (4na,2nh). Most coefficients have a non-zero value, showing a high richness of Fourier modes.*

5 Discussion

As expected, the exponential encoding can model a frequency range more than large enough to encapsulate the toy problem. The higher error found on the sphere problem is due in part to the far larger output range. Even considering this the error is still relatively large which can likely be attributed to the function's lack of periodicity considering its smaller scaling. The good performances on the Gaussian prove that its frequency range is smaller than exponential. It is probably constant but that would have to be confirmed through further experimentation.

The differences in performances between Pauli re-uploading and exponential encoding are as expected because of the difference in their frequency range scaling. However, the underperformance of quadratic encoding is more unexpected. The two possibilities currently considered are that although slower, the linear scaling can better model the exponential scaling of frequency ranges. The other possibility is that this encoding had unexpected consequences on the interval between encoded frequencies, preventing it from properly modelling the function. However, further experiments are necessary to reach a definitive conclusion.

The two reservoirs performed quite similarly with a slight advantage to the Haar random unitary as was expected from the results presented in Xiong *et al.* [3]. However, it should be kept in mind that the chaotic Ising would be preferable in practice, as it would be less likely to create an exponential concentration of observables' expected values.

The classification results suggest the preprocessing was effective, achieving almost 50% accuracy on 10 classes with only 2 scalars. Additionally, further tests with larger circuits only lowered the accuracy. Combined with the encoding used, this indicates that these scalars have approximately 4 frequencies each, as they are best represented using 2 accessible qubits and exponential encoding, which also shows the wide applicability of exponential encoding. Finally, the wide array of non-zero Fourier coefficients indicates that the reservoir fulfilled its scrambling role properly.

For the experiments, running the QELM was simulated by directly calculating the expectation values from the provided circuit, as this results in the most accurate output values and computes faster. However, this is impractical on real quantum hardware, where the expectation values need to be derived through averaging outputs across several repetitions. This will likely introduce some noise in the system, impacting performance, as well as inducing the exponential concentration mentioned previously.

Regarding the classification task, several other methods were considered for preprocessing the input data. Most notably, Principal Component Analysis (PCA) could have been used to reduce the dimensionality of the input state vector. Then, the components can be used as the new input vector. However, due to the low number of accessible qubits that were tested with ($n_a < 10$), PCA would capture too few components to provide meaningful information about the image. Therefore, it has not been included in the preprocessing section. Future research could investigate the effect that PCA and other preprocessing methods have on the performance of QELM image classification.

6 Conclusion

This report has investigated the potential of QELMs on a variety of tasks, including the prediction of the Fourier series of several Regression functions and classification. It showcases the performances of various encoding schemes and reservoirs when applied to different target functions with varying frequency spectra. The smallest of these proved to be the toy function, with an MSE of $1.44 * 10^{-11}$. Furthermore, it was shown an encoding scheme encompassing exponentially many frequencies and a rich reservoir model provide highly accurate results on this task. Finally, the classification results show a performance of around 50% over 10 different classes, showing the versatility of QELMs for various different prediction tasks.

References

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